

```
C:\WINDOWS\system32\cmd. x + v
(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-03>notepad app\preprocessing.py

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-03>python -m app.data_loader
=== OmniRetail AI Checkpoint 3 Data Ingestion ===

Clickstream: (1000, 16)
Product Reviews: (800, 11)
E-commerce Test Cases: 50

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-03>python -c "from app.data_loader import load_clickstream; from app.preprocessing import clean_clickstream, prepare_features; df=clean_clickstream(load_clickstream()); X,y,p=prepare_features(df); print('Cleaned data:', df.shape); print('Features:', X.shape); print('Target:', y.shape); print('Preprocessing: PASSED')"
Cleaned data: (1000, 16)
Features: (1000, 9)
Target: (1000,)
Preprocessing: PASSED

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-03>notepad app\core_model.py

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-03>python -m app.core_model
Loading clickstream dataset...
Original dataset shape: (1000, 16)
Cleaned dataset shape: (1000, 16)
Feature records: (1000, 9)
Target records: (1000,)
Training records: 800
Testing records: 200
2026/10/05 19:28:14 INFO mlflow.store.db.utils: Creating initial MLflow database tables...
2026/10/05 19:28:14 INFO mlflow.store.db.utils: Updating database tables
2026/10/05 19:28:18 INFO mlflow.tracking.fluent: Experiment with name 'OmniRetail_Checkpoint_3_Core_Model' does not exist. Creating a new experiment.

=====
OmniRetail AI Core Model Experiments
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```

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C:\WINDOWS\system32\cmd. x + v

=====
OmniRetail AI Core Model Experiments
=====

Running: RandomForest_Config1
Accuracy : 0.5350
Precision: 0.5489
Recall : 0.9099
F1 Score : 0.6847
ROC-AUC : 0.4258

Running: RandomForest_Config2
Accuracy : 0.5100
Precision: 0.5430
Recall : 0.7387
F1 Score : 0.6260
ROC-AUC : 0.4307

Running: RandomForest_Config3
Accuracy : 0.4850
Precision: 0.5290
Recall : 0.6577
F1 Score : 0.5863
ROC-AUC : 0.4427

=====
BEST CONFIGURATION
=====
Configuration : RandomForest_Config1
Accuracy : 0.5350
Precision : 0.5489
Recall : 0.9099
F1 Score : 0.6847
ROC-AUC : 0.4258

Best model saved : C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-03\models\best_core_model.jobl
```

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C:\WINDOWS\system32\cmd. x + v

Best model saved : C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-03\models\best_core_model.joblib
Results saved : C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-03\docs\experiment_results.json
MLflow database : C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-03\mlflow.db

Core model experimentation completed successfully.

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-03>type docs\experiment_results.json
{
  "experiment": "OmniRetail_Checkpoint_3_Core_Model",
  "dataset": "aiml_ecommerce_clickstream.csv",
  "total_records": 1000,
  "training_records": 800,
  "testing_records": 200,
  "feature_count": 9,
  "target": "label_purchased",
  "selection_metric": "f1_score",
  "experiments_completed": 3,
  "best_configuration": {
    "configuration": "RandomForest_Config_1",
    "model": "Random Forest",
    "n_estimators": 100,
    "max_depth": 5,
    "min_samples_split": 2,
    "accuracy": 0.535,
    "precision": 0.5489,
    "recall": 0.9099,
    "f1_score": 0.6847,
    "roc_auc": 0.4258
  },
  "all_results": [
    {
      "configuration": "RandomForest_Config_1",
      "model": "Random Forest",
      "n_estimators": 100,
      "max_depth": 5,
      "min_samples_split": 2,
      "accuracy": 0.535,
      "precision": 0.5489,
      "recall": 0.9099,
      "f1_score": 0.6847,
      "roc_auc": 0.4258
    },
    {
      "configuration": "RandomForest_Config_2",
      "model": "Random Forest",
      "n_estimators": 200,
      "max_depth": 10,
      "min_samples_split": 2,
      "accuracy": 0.51,
      "precision": 0.543,
      "recall": 0.7387,
      "f1_score": 0.626,
      "roc_auc": 0.4307
    },
    {
      "configuration": "RandomForest_Config_3",
      "model": "Random Forest",
      "n_estimators": 300,
      "max_depth": null,
      "min_samples_split": 2,
      "accuracy": 0.485,
      "precision": 0.529,
      "recall": 0.6577,
      "f1_score": 0.5863,
      "roc_auc": 0.4427
    }
  ]
}
```

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    "n_estimators": 100,
    "max_depth": 5,
    "min_samples_split": 2,
    "accuracy": 0.535,
    "precision": 0.5489,
    "recall": 0.9099,
    "f1_score": 0.6847,
    "roc_auc": 0.4258
  },
  {
    "configuration": "RandomForest_Config_2",
    "model": "Random Forest",
    "n_estimators": 200,
    "max_depth": 10,
    "min_samples_split": 2,
    "accuracy": 0.51,
    "precision": 0.543,
    "recall": 0.7387,
    "f1_score": 0.626,
    "roc_auc": 0.4307
  },
  {
    "configuration": "RandomForest_Config_3",
    "model": "Random Forest",
    "n_estimators": 300,
    "max_depth": null,
    "min_samples_split": 2,
    "accuracy": 0.485,
    "precision": 0.529,
    "recall": 0.6577,
    "f1_score": 0.5863,
    "roc_auc": 0.4427
  }
]
}

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-03>python -c "import joblib; model=joblib.load('models\best_core_model.joblib'); print('Loaded model:', type(model).__name__); print('Best model artifact: PASSED')"
```

```
C:\WINDOWS\system32\cmd. x + v

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-03>python -c "import joblib; model=joblib.l
oad('models/best_core_model.joblib'); print('Loaded model:', type(model).__name__); print('Best model artifact: PASSED')"
Loaded model: Pipeline
Best model artifact: PASSED

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-03>type docs\experiment_results.json
{
  "experiment": "OmniRetail_Checkpoint_3_Core_Model",
  "dataset": "aiml_ecommerce_clickstream.csv",
  "total_records": 1000,
  "training_records": 800,
  "testing_records": 200,
  "feature_count": 9,
  "target": "label_purchased",
  "selection_metric": "f1_score",
  "experiments_completed": 3,
  "best_configuration": {
    "configuration": "RandomForest_Config_1",
    "model": "Random Forest",
    "n_estimators": 100,
    "max_depth": 5,
    "min_samples_split": 2,
    "accuracy": 0.535,
    "precision": 0.5489,
    "recall": 0.9099,
    "f1_score": 0.6847,
    "roc_auc": 0.4258
  },
  "all_results": [
    {
      "configuration": "RandomForest_Config_1",
      "model": "Random Forest",
      "n_estimators": 100,
      "max_depth": 5,
      "min_samples_split": 2,
      "accuracy": 0.535,
```

```
      "max_depth": 5,
      "min_samples_split": 2,
      "accuracy": 0.535,
      "precision": 0.5489,
      "recall": 0.9099,
      "f1_score": 0.6847,
      "roc_auc": 0.4258
    },
    {
      "configuration": "RandomForest_Config_2",
      "model": "Random Forest",
      "n_estimators": 200,
      "max_depth": 10,
      "min_samples_split": 2,
      "accuracy": 0.51,
      "precision": 0.543,
      "recall": 0.7387,
      "f1_score": 0.626,
      "roc_auc": 0.4307
    },
    {
      "configuration": "RandomForest_Config_3",
      "model": "Random Forest",
      "n_estimators": 300,
      "max_depth": null,
      "min_samples_split": 2,
      "accuracy": 0.485,
      "precision": 0.529,
      "recall": 0.6577,
      "f1_score": 0.5863,
      "roc_auc": 0.4427
    }
  ]
}

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-03>notepad tests\test_core_model.py

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-03>pytest -q
```

```
C:\WINDOWS\system32\cmd. x + v

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-03>notepad tests\test_core_model.py

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-03>pytest -q
..... [100%]
8 passed in 17.31s

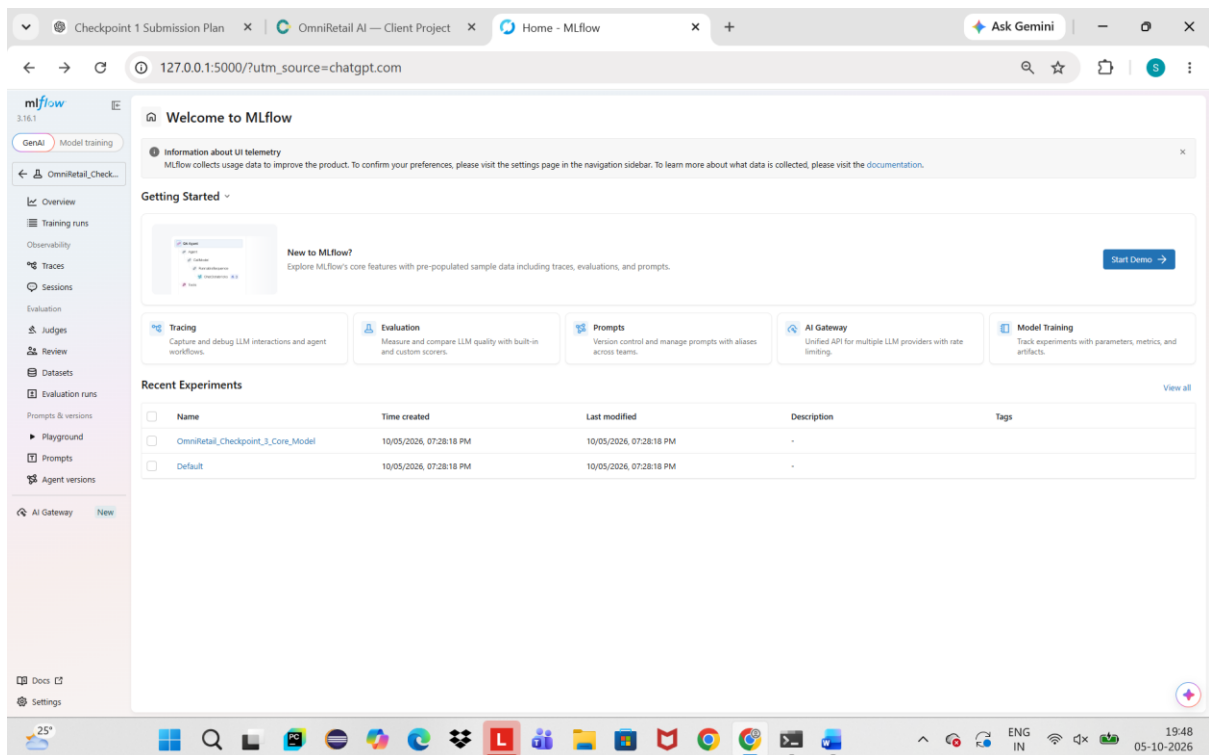
(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-03>notepad docs\CORE_MODEL_REPORT.md

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-03>dir docs
Volume in drive C is Windows-SSD
Volume Serial Number is EA8D-9B22

Directory of C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-03\docs

05-10-2026 19:42 <DIR> .
05-10-2026 19:40 <DIR> ..
05-10-2026 19:42          5,278 CORE_MODEL_REPORT.md
05-10-2026 19:31       1,841 experiment_results.json
                2 File(s)          7,119 bytes
                2 Dir(s)  9,100,816,384 bytes free

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-03>mlflow ui --backend-store-uri sqlite:///
mlflow.db --port 5000
Registry store URI not provided. Using backend store URI.
[MLflow] Security middleware enabled with default settings (localhost-only). To allow connections from other hosts, use --host 0.0.0.
0 and configure --allowed-hosts and --cors-allowed-origins.
2026/10/05 19:43:36 WARNING mlflow.server: MLflow job execution requirements not met (MLflow job backend does not support Windows sys
tem.). Server will start without job execution support. Errors will be surfaced at job invocation time.
2026/10/05 19:43:36 INFO: Uvicorn running on http://127.0.0.1:5000 (Press CTRL+C to quit)
2026/10/05 19:43:36 INFO: Started parent process [3644]
C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-03\.venv\Lib\site-packages\mlflow\server\fastapi_ap
p.py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future release. Please refer t
o https://github.com/abersheeran/a2wsgi as a replacement.
  from starlette.middleware.wsgi import WSGIResponder, build_enviro
C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-03\.venv\Lib\site-packages\mlflow\server\fastapi_ap
p.py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future release. Please refer t
```



Checkpoint 1 Submission Plan x OmniRetail AI — Client Project x Runs - Experiment 1 - MLflow x + Ask Gemini

127.0.0.1:5000/?utm_source=chatgpt.com#/experiments/1/runs?workflowType=genai&searchFilter=&orderByKey=attributes.start_time&orderBy...

mlflow 3.16.1

GenAI Model training

OmniRetail_Check...

Overview Training runs Observability Traces Sessions Evaluation Judges Review Datasets Evaluation runs Prompts & versions Playground Prompts Agent versions AI Gateway New

Experiments > OmniRetail_Checkpoint_3_Core_Model >

Training runs

Q metrics.rmse < 1 and params.model = "tree" All time State: Active Datasets Sort: Created Columns Group by New run

	Run Name	Created	Dataset	Duration	Source	Models	Metrics
☐	RandomForest_Config_3	18 minutes ago	-	16.0s	C:\Users...	core_random_forest_model	accuracy 0.485
☐	RandomForest_Config_2	18 minutes ago	-	14.8s	C:\Users...	core_random_forest_model	0.51
☐	RandomForest_Config_1	18 minutes ago	-	21.6s	C:\Users...	core_random_forest_model	0.535
☐	RandomForest_Config_1	20 minutes ago	-	3.7s	C:\Users...	core_random_forest_model	0.535

Show more columns (13 total)

4 matching runs

25° Windows Search File Explorer Edge VS Code L Docker Desktop Telegram OneDrive Outlook Mail Google Chrome Task View Settings

ENG IN 19:49 05-10-2026

Checkpoint 1 Submission Plan x OmniRetail AI — Client Project x Run ae59a58ed43641e0aeeb0e72fd0090d5 + Ask Gemini

127.0.0.1:5000/?utm_source=chatgpt.com#/experiments/1/runs/ae59a58ed43641e0aeeb0e72fd0090d5

mlflow 3.16.1

GenAI Model training

OmniRetail_Check...

Overview Training runs Observability Traces Sessions Evaluation Judges Review Datasets Evaluation runs Prompts & versions Playground Prompts Agent versions AI Gateway New

OmniRetail_Checkpoint_3_Core_Model > Runs >

RandomForest_Config_2

Overview Model metrics System metrics Traces Artifacts

Metrics (5)

Q Search metrics

Metric	Value	Models
accuracy	0.51	core_random_forest_model
precision	0.543046357615894	core_random_forest_model
recall	0.7387387387387387	core_random_forest_model
f1_score	0.6259541984732825	core_random_forest_model
roc_auc	0.43071161048689144	core_random_forest_model

Parameters (6)

Q Search parameters

Parameter	Value
model	Random Forest
n_estimators	200
max_depth	10
min_samples_split	2
random_state	42
test_size	0.2

Logged models (1)

Type	Step	Model name	Status	Created	Registered models	Dataset	accuracy
Output	0	core_random_forest	Ready	18 minutes ago	-	-	0.51

Model attributes No dataset

Unassigned Experiment ID 1 Status Finished Run ID ae59a58ed43641e0aeeb0e72fd0090d5 Duration 14.8s Source C:\Users\Sanjana C:\OneDrive\Documents\OmniRetail\OmniRetail_AI_Checkpoint_03\app\core_model.py Registered prompts — Datasets None Tags Add tags Registered models None

25° Windows Search File Explorer Edge VS Code L Docker Desktop Telegram OneDrive Outlook Mail Google Chrome Task View Settings

ENG IN 19:49 05-10-2026